INTELIGÊNCIA ARTIFICIAL

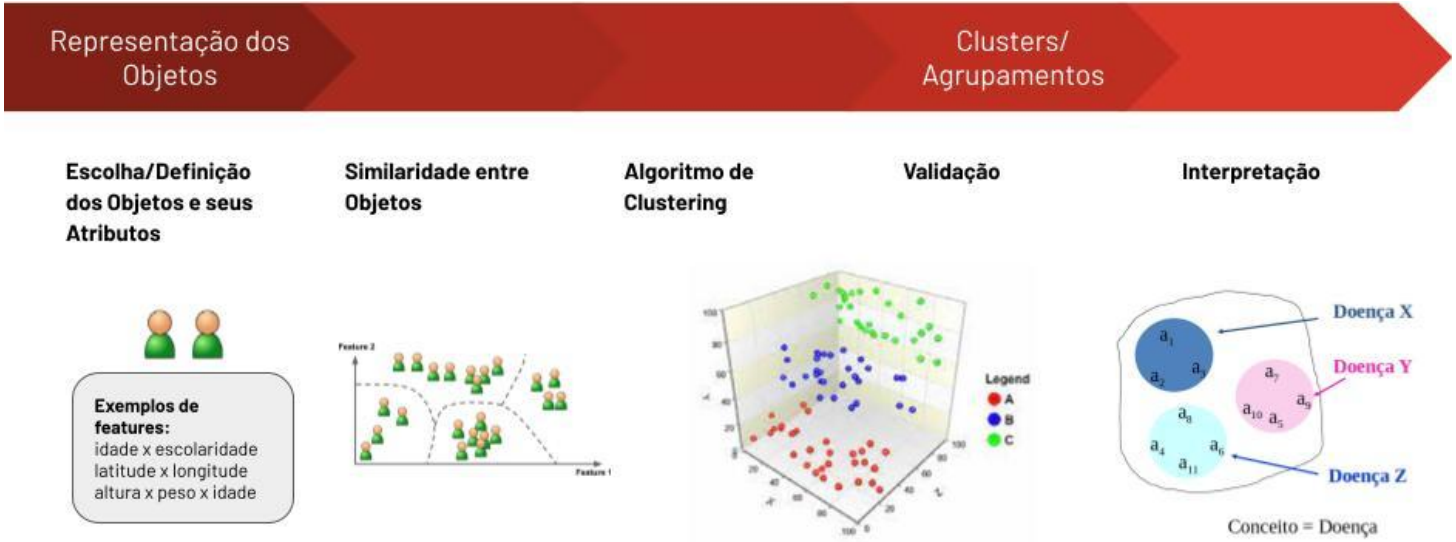
Aprendizado Descritivo
Clustering

Marcos de Souza
mso2@cesar.school

C.E.S.A.R school

Aprendizado Não-Supervisionado (Descritivo)

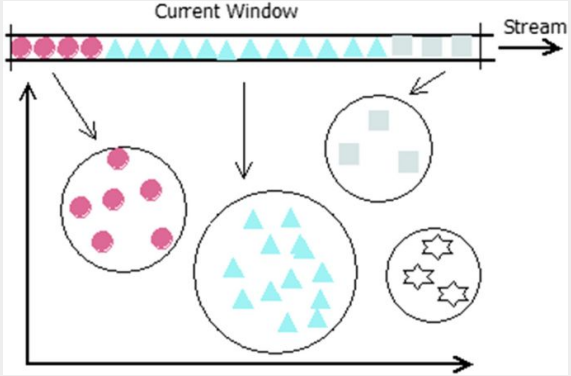
O resultado normalmente são agrupamentos (clusters) dos exemplos da base de treinamento que podem ser utilizados para predizer o grupo de um novo indivíduo.



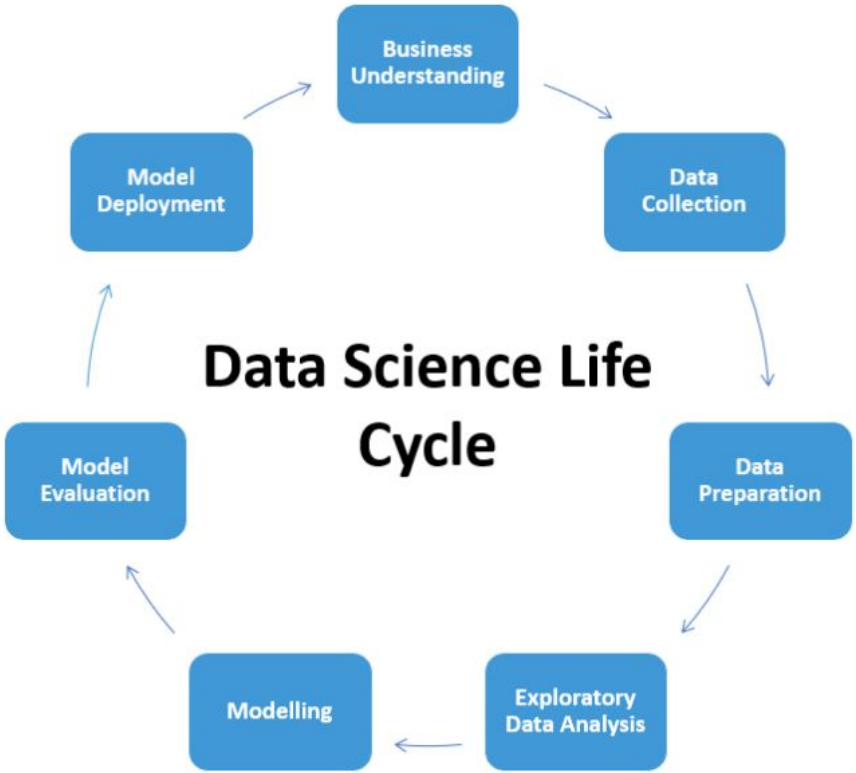
Obs: a normalização não altera o arranjo natural.

Data Pipeline

Ref: https://laptrinhx.com/complete-life-cycle-of-a-data-science-machine-learning-project-3256392522/



Namitha, K., Santhosh Kumar, G. Learning in the presence of concept recurrence in data stream clustering. *J Big Data* 7, 75 (2020).



mas antes …

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Avaliação de Agrupamentos

Índices Internos (não supervisionado)

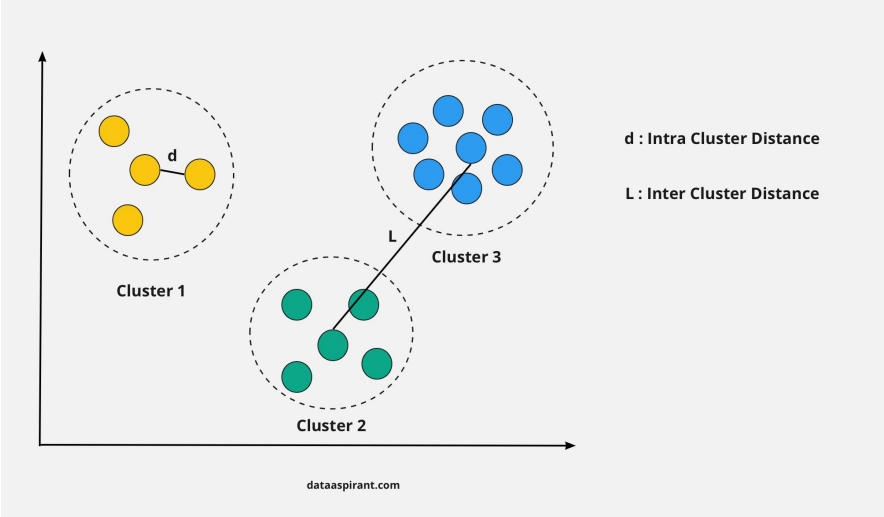
Avaliação de Agrupamentos

Principais características

- Alta densidade **em um mesmo** grupo (homogeneidade)
- Alta separabilidade **entre** grupos (heterogeneidade)

Outras características

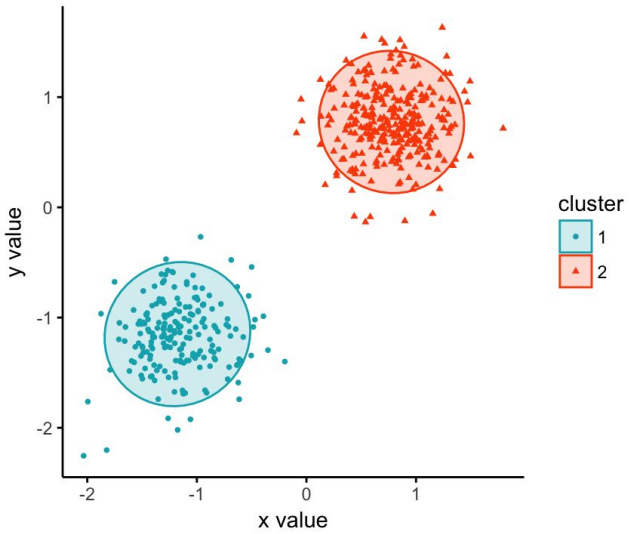
- Balanceamento de itens por grupo

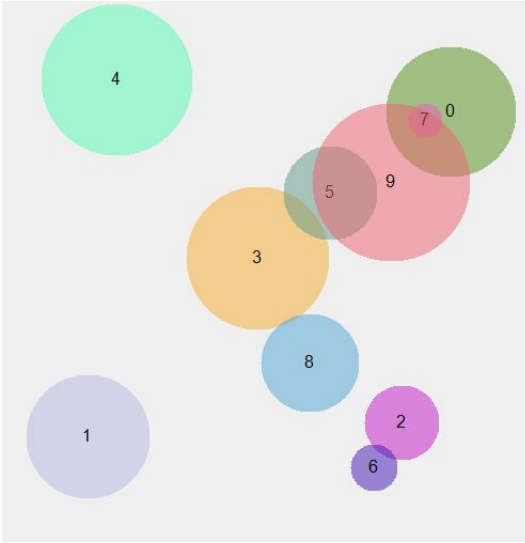


O que seria um “bom” agrupamento ?

Um resultado esperado de uma atividade de clustering é a geração de grupos

- com **alta densidade** (menor distância entre os pontos de um mesmo grupo)
- e **maior separabilidade** (maior distância entre os grupos).





Inertia

number of clusters

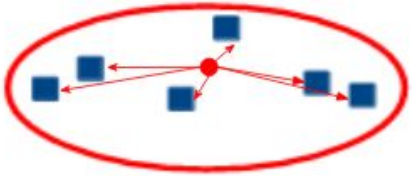
number of cases

centroid for cluster *j*

case *i*

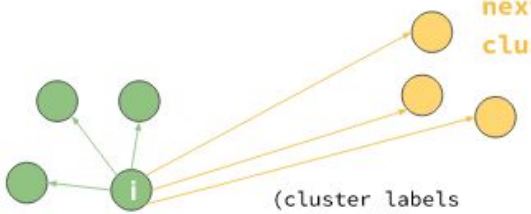
objective function ←

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} \underbrace{\left\| x_i^{(j)} - c_j \right\|^2}_{\text{Distance function}}$$



Intra cluster distance

Silhouette



b(i): distance between i and its **next nearest cluster centroid**

(cluster labels assigned from ground-truth classes)

a(i): average distance between i and all of the other points in **its own cluster**

For a single point, i

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$$-1 \leq s(i) \leq 1$$

Sprawling, overlapped clusters

Tight, well-separated clusters

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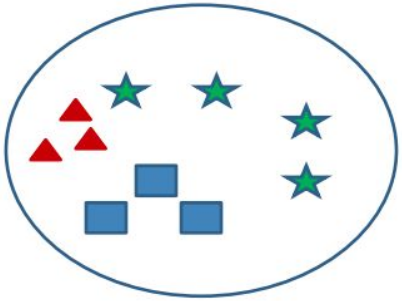
Avaliação de Agrupamentos

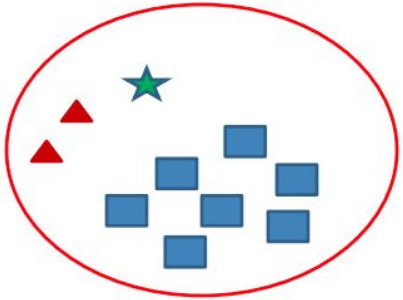
Índices Externos (supervisionado)

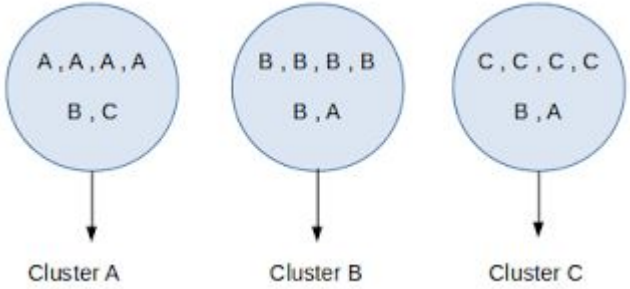
Normalized Mutual Information (NMI)

ref: https://course.ccs.neu.edu/cs6140sp15/7_locality_cluster/Assignment-6/NMI.pdf

- Quanto maior (max=1) melhor.
- Avalia o quão "puro" são os grupos (baixa entropia)







Normalized Mutual Information (NMI)

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$$NMI(Y,C) = \frac{2 \times I(Y;C)}{[H(Y) + H(C)]}$$

- wher-,

1) Y = class labels

2) C = cluster labels

3) H(.) = Entropy

4) I(Y;C) = Mutual Information b/w Y and C

Note: All logs are base-2.

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$$H(i) = -\sum_{j \in K} p(i_j) \log_2 p(i_j)$$

Where $p(i_j)$ is the probability of a point in the cluster i of being classified as class j. For instance, if you have 10 points in cluster i and based on the labels of your true data you have 6 in class A, 3 in class B and 1 in class C. Then your entropy of cluster i will be:

$$H(i) = -\left(\frac{6}{10}\log\frac{6}{10} + \frac{3}{10}\log\frac{3}{10} + \frac{1}{10}\log\frac{1}{10}\right)$$

Once you have the entropy of each cluster, the overall entropy is just the weighted sum of the entropies of each cluster. You can compute the overall entropy using the following formula:

$$H = \sum_{i \in C} H(i)\frac{N_i}{N}$$

where H is the entropy, N_i is the number of points in the cluster i and N is the total number of points.

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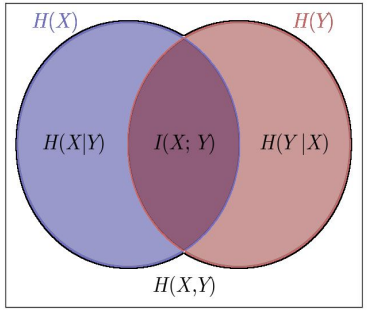
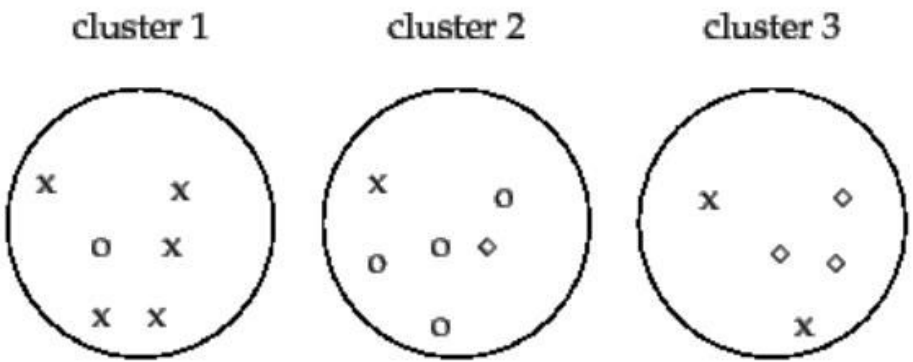
$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P(x, y) \log_2 \left(\frac{P(x, y)}{P(x) P(y)} \right)$$



$$\begin{aligned} \mathrm{I}(X;Y) &\equiv \mathrm{H}(X) - \mathrm{H}(X \mid Y) \\ &\equiv \mathrm{H}(Y) - \mathrm{H}(Y \mid X) \\ &\equiv \mathrm{H}(X) + \mathrm{H}(Y) - \mathrm{H}(X,Y) \\ &\equiv \mathrm{H}(X,Y) - \mathrm{H}(X \mid Y) - \mathrm{H}(Y \mid X) \end{aligned}$$

Purity



$$Purity = \frac{1}{k}\sum_{r=1}^{k} P(C_r) = \frac{1}{17} \times (5+4+3) = \frac{12}{17}$$ **0.71**

Code

```
# importa o pacote cluster
from sklearn import metrics

# silhouette
sil = metrics.silhouette_score(X, labels, metric='euclidean')

# NMI
nmi = metrics.normalized_mutual_info_score(y_true, labels)
```

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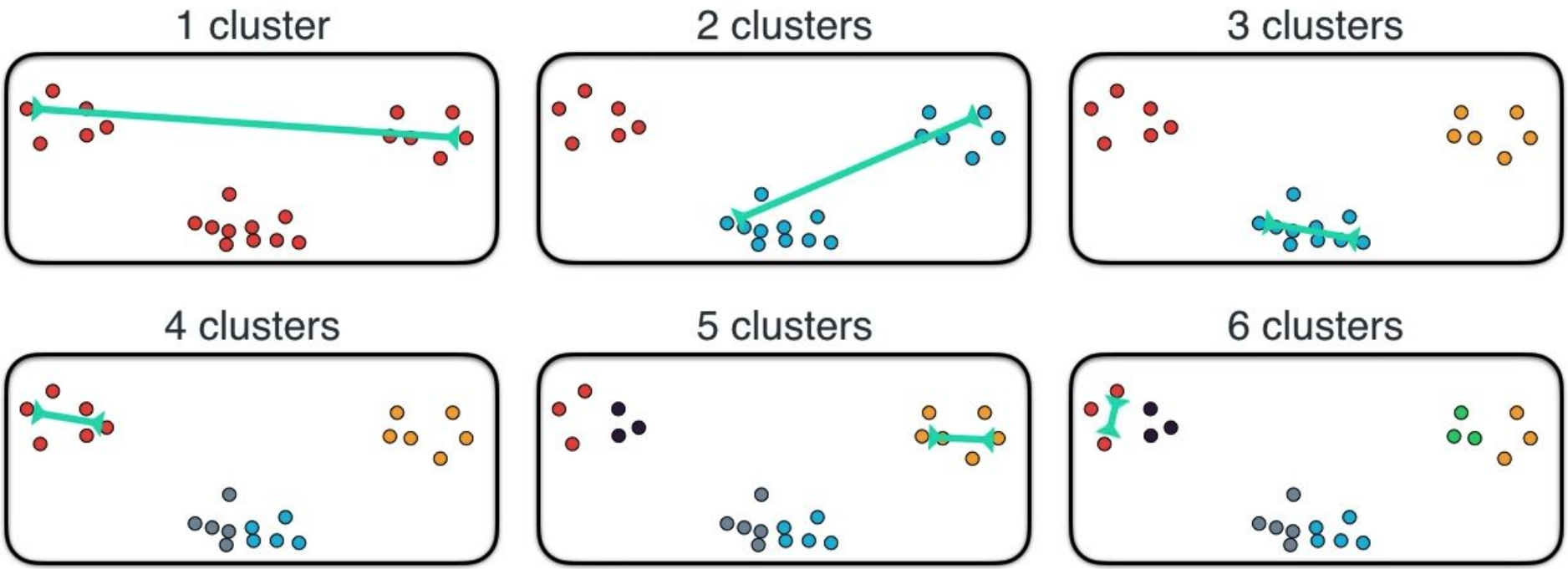
Elbow Method

Utiliza-se índices internos na tentativa de identificar o melhor número de clusters

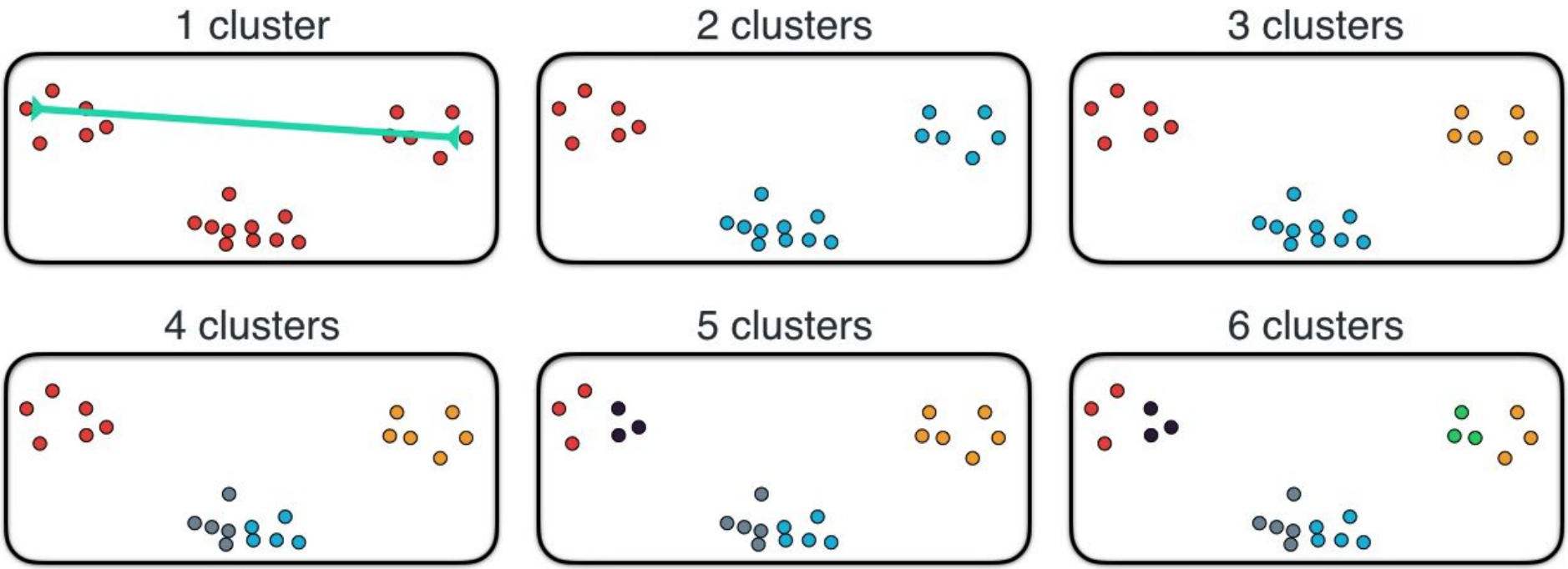
Como definir k?

Calcular o K-Means para diversos valores de k



Como definir k?

Calcular a maior distância entre exemplos de um mesmo grupo



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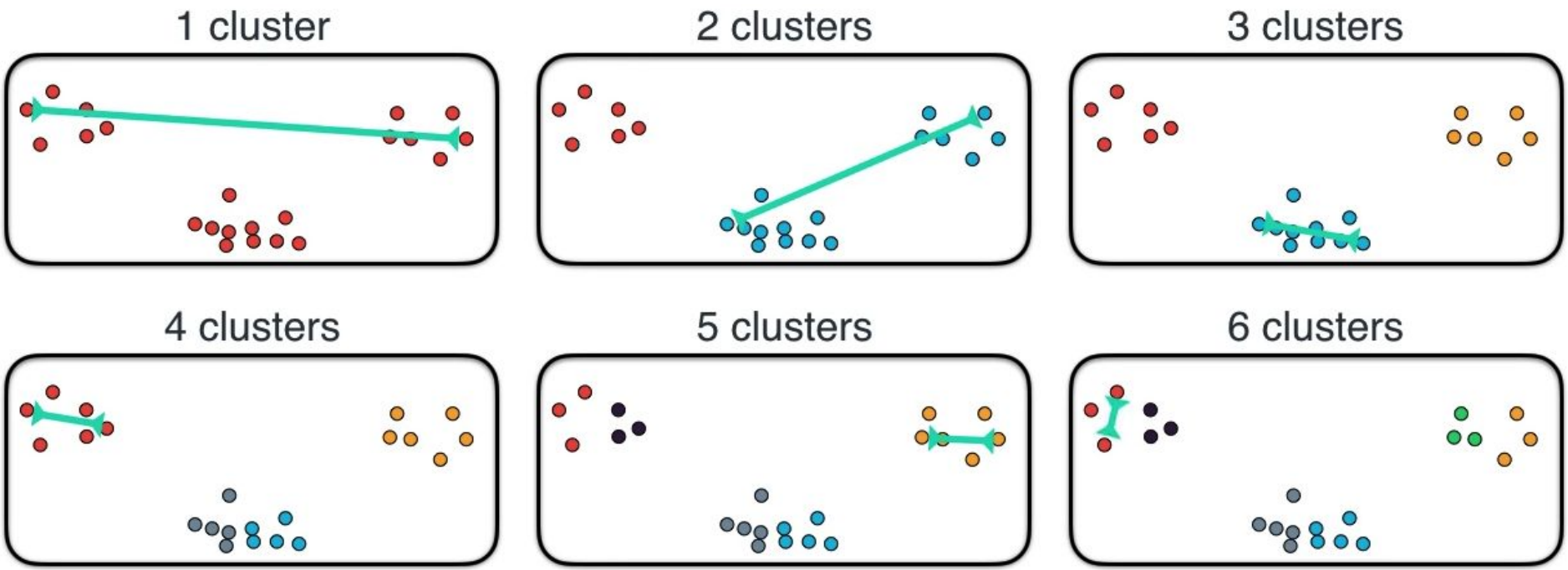
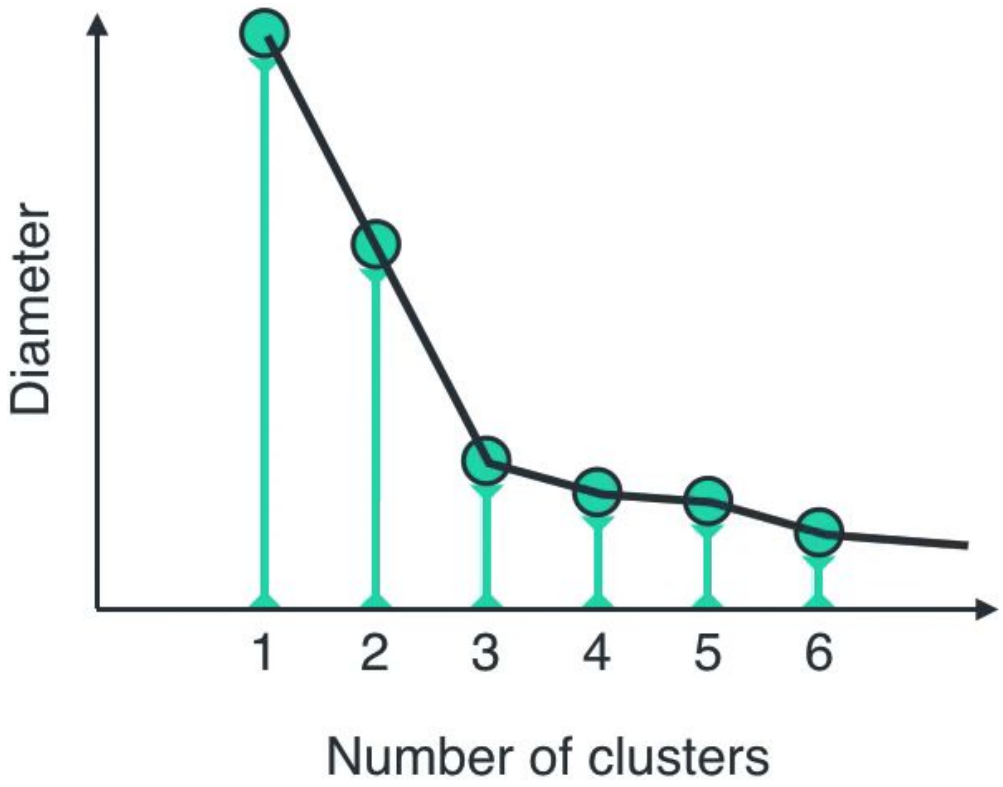


Como definir k?

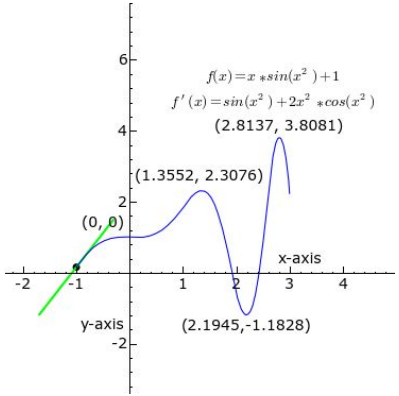
Plotar a variação intra-clusters pelo número de clusters (k).

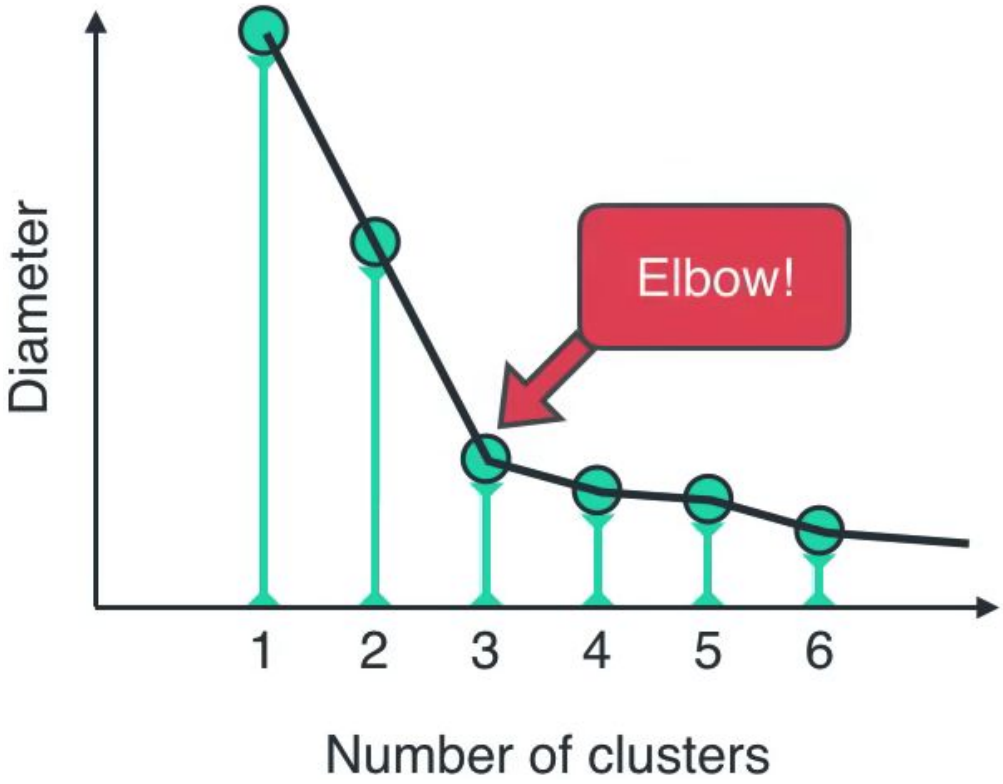
Procurar pelo 'cotovelo' da curva!

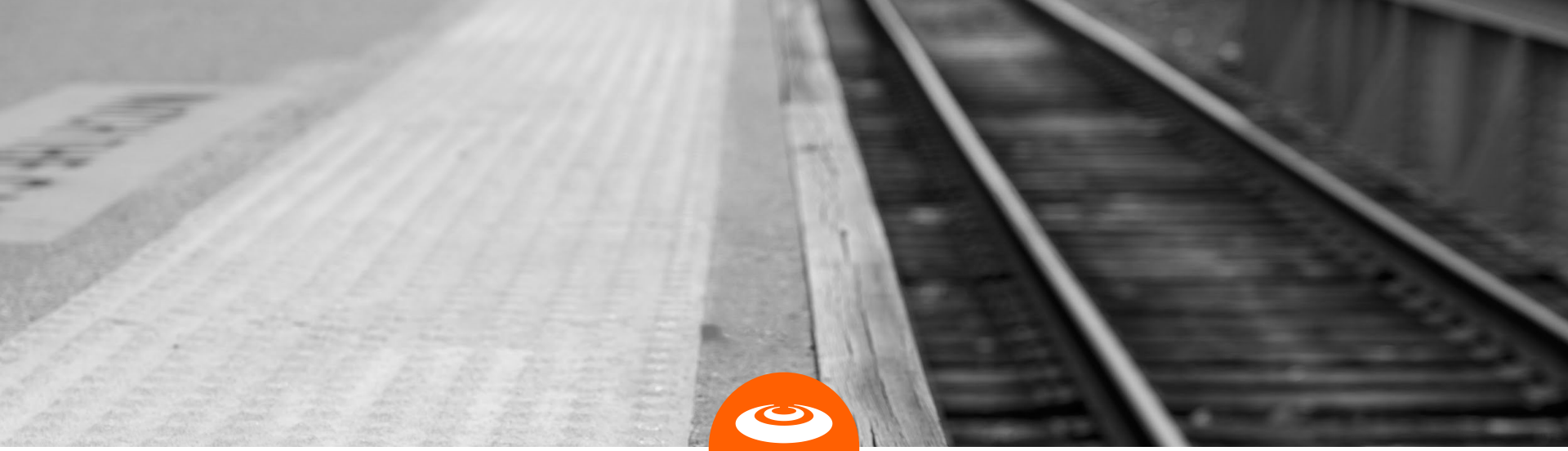


Como definir k?

O melhor valor de k será definido pelo ponto no qual um aumento no número de clusters não significa uma grande diminuição no diâmetro.







Hands On!

Experimentos em Machine Learning com Python

Hands On

- Nesta atividade iremos utilizar os datasets da atividade anterior
- Iremos aplicar os diversos algoritmo de clustering
- Iremos calcular as métricas e discutir os resultados
- Iremos tentar detectar o número de grupos

C . e . S . A . R

Pessoas impulsionando inovação.

Inovação impulsionando negócios.

NOSSO CONTATO

mso@cesar.org.br
mso2@cesar.school